Aim:

To integrate JavaScript with Java Servlets in order to implement a dynamic shopping cart where items and their quantities are updated instantly without reloading the webpage.

Description:

This experiment illustrates the interaction between frontend JavaScript and backend Java Servlets to build a responsive shopping cart. The user interface is created using HTML and JavaScript, where users can add products using buttons.

Instead of refreshing the page for every action, JavaScript sends asynchronous requests to the servlet using **XMLHttpRequest (AJAX)**. This allows smooth communication between client and server without interrupting the user experience.

On the server side, the CartServlet processes incoming requests and stores cart data using **HTTP Session**. Each product's quantity is updated in a HashMap stored in the session. The servlet then sends the updated cart details (product IDs and quantities) back to the client.

The JavaScript receives this response and dynamically updates the webpage to reflect the latest cart contents. This demonstrates how AJAX enables real-time updates and improves efficiency in web applications.

Steps to Execute:

1. Open Eclipse IDE
   - Launch Eclipse
   - Select your workspace

2. Create Dynamic Web Project
   - Navigate to:
     File → New → Dynamic Web Project
   - Enter project name: ShoppingCartApp
   - Select Apache Tomcat v9.0
   - Click **Finish**

3. Create Package
   - Go to src/main/java
   - Right-click → New → Package

- Name:

  com.example.servlet

4. Create Servlet
   - Right-click package → New → Servlet
   - Name: CartServlet
   - Add servlet mapping:

     @WebServlet("/cart")
   - Implement logic to:
     - Retrieve session
     - Store cart using HashMap
     - Update product quantity
     - Send updated cart data as response

5. Create HTML File
   - Right-click src/main/webapp
   - New → HTML File
   - Name: index.html
   - Add:
     - Product buttons
     - Cart display section (<p id="status">)

6. Create JavaScript File
   - Inside webapp, create folder: js
   - Inside it create file: script.js
   - Implement:
     - AJAX using XMLHttpRequest
     - Send product ID and quantity to servlet
     - Receive updated cart data
     - Dynamically update UI

7. Link JavaScript to HTML
Add inside index.html:

```
<script src="js/script.js"></script>
```

8. Configure Server
   - Open **Servers tab**
   - Right-click → New → Server
   - Select Apache Tomcat v9.0
   - Add your project

9. Run the Project
- Right-click project
- Select:
  Run As → Run on Server

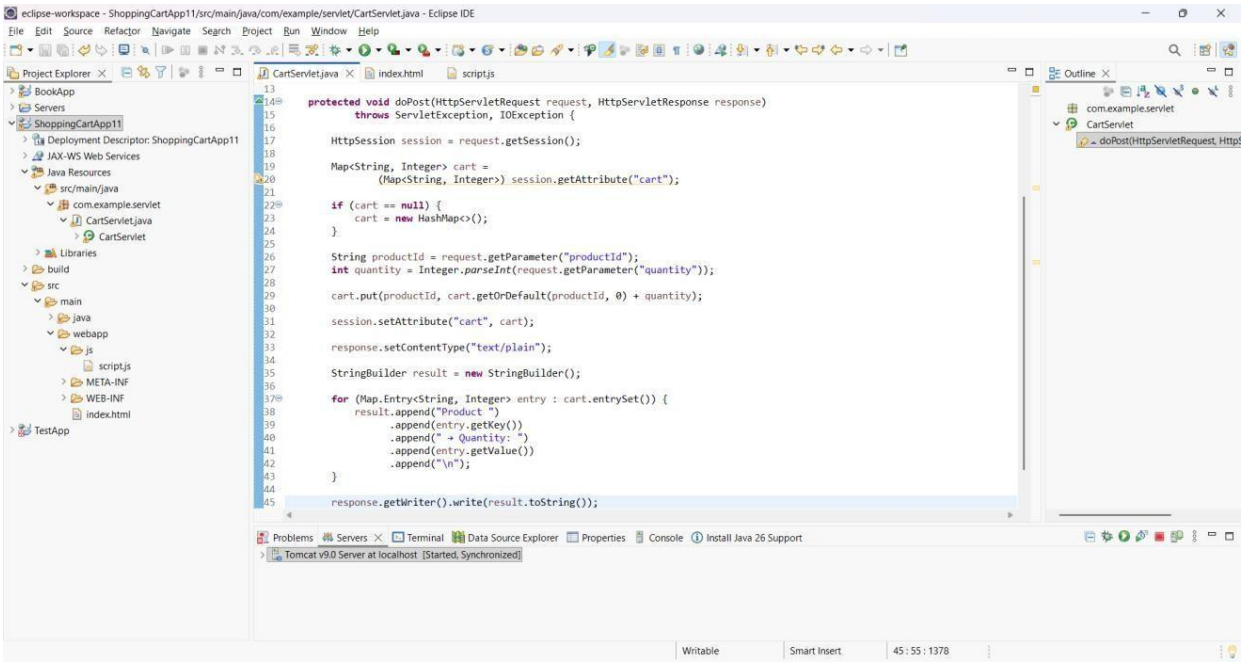
10. Open Application in Browser
http://localhost:8080/ShoppingCartApp/

11. Test Functionality
- Click:
  - Add Product 101
  - Add Product 102
- Observe:
  - Page does NOT refresh
  - Cart updates instantly
  - Quantities increase dynamically

12. Verify Output
Initial:

(no items displayed)

File Structure :



Code Implementation

1. **CartServlet.java**

```
package com.example.servlet;

import java.io.IOException;
import java.util.HashMap;
import java.util.Map;

import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/cart")
public class CartServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        HttpSession session = request.getSession();

        Map<String, Integer> cart =
                (Map<String, Integer>) session.getAttribute("cart");

        if (cart == null) {
            cart = new HashMap<>();
        }

        String productId = request.getParameter("productId");
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        cart.put(productId, cart.getOrDefault(productId, 0) + quantity);

        session.setAttribute("cart", cart);

        response.setContentType("text/plain");

        StringBuilder result = new StringBuilder();

        for (Map.Entry<String, Integer> entry : cart.entrySet()) {
            result.append("Product ")
                    .append(entry.getKey())
                    .append(" → Quantity: ")
                    .append(entry.getValue())
                    .append("\n");
```

```
}

        response.getWriter().write(result.toString());
        response.getWriter().write("Item added to cart");
    }
}
```

2. script.js

```
function addToCart(productId) {

    var xhr = new XMLHttpRequest();

    xhr.open("POST", "cart", true);

    xhr.setRequestHeader("Content-Type",
        "application/x-www-form-urlencoded");

    xhr.onload = function () {
        if (xhr.status === 200) {

            let data = xhr.responseText.split("\n");

            let output = "";

            data.forEach(line => {
                if (line.trim() !== "") {
                    output += "<p>" + line + "</p>";
                }
            });

            document.getElementById("status").innerHTML = output;
        }
    };

    xhr.send("productId=" + productId + "&quantity=1");
}
```

3. cart.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Shopping Cart</title>
</head>
<body>

<h2>Products</h2>

<button onclick="addToCart('101')">Add Product 101</button>
<button onclick="addToCart('102')">Add Product 102</button>

<p id="status"></p>

<script src="js/script.js"></script>

</body>
</html>
```

Output:

Shopping Cart
localhost:8080/ShoppingCartApp11/
23071A05H5
Products
Add Product 101
Add Product 102